Prob. 2.

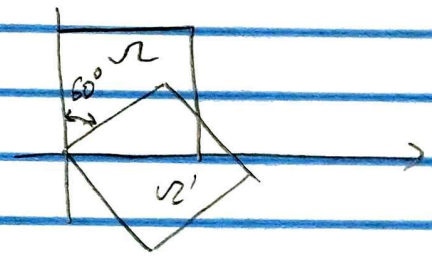
Show that if $u(x,y)$ is a harm-c fn in Ω, then $u(x',y')$ is harmonic in the domain where

(1) $$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(2) $$\Omega' = \{ (\cos(\alpha)x + y\sin(\alpha),\ -\sin(\alpha)x + \cos(\alpha)y) : (x,y) \in \Omega \}$$

A sample Ω & Ω' for $\alpha = \pi/3$ are given as:

(3)



sol-n.

By the instructions given first, we need to check 2 things

(i) $|J| \neq 1$ ((1) is well-defined)

(ii) Check $u_{x'x'} + u_{y'y'}$ stays the same.

Of course the tr-n in (1) is well defined:

(i) $$J = \begin{vmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{vmatrix} = \cos^2\alpha + \sin^2\alpha = 1 \neq 0 \Rightarrow$$

(1) is OK.

(ii) Let (4) $$\begin{cases} \xi = \cos\alpha \cdot x + \sin\alpha \cdot y \\ \eta = -\sin\alpha\, x + \cos\alpha \cdot y \end{cases} \Rightarrow$$ Then

$\xi = \cos\alpha\, x + \sin\alpha\, y$
$\eta = -\sin\alpha\, x + \cos\alpha\, y$

(5) $u_{xx} = u_{\xi\xi} \cdot \xi_x^2 + 2u_{\xi\eta}\, \xi_x \eta_x + u_{\eta\eta} (\eta_x)^2 + u_\xi\, \xi_{xx} + u_\eta\, \eta_{xx} =$

$= u_{\xi\xi} (\cos\alpha)^2 + 2u_{\xi\eta} (\cos\alpha)\cdot(-\sin\alpha) + u_{\eta\eta} \cdot (-\sin\alpha)^2 + 0$ &

(6) $u_{yy} = u_\xi \cdot (\sin\alpha)^2 + 2u_{\xi\eta} (-\sin\alpha)\cdot(\cos\alpha) + u_{\eta\eta} \cdot (\cos\alpha)^2$. Then

Laplace eq-n, by (5) & (6):

$u_{xx} + u_{yy} = u_{\xi\xi} (\cos^2\alpha + \sin^2\alpha) + 2u_{\xi\eta} \overbrace{(-\cos\alpha\sin\alpha + \sin\alpha\cos\alpha)}^{=0} +$

$+ u_{\eta\eta} (\sin^2\alpha + \cos^2\alpha) = u_{\xi\xi} + u_{\eta\eta} = 0 \Rightarrow$

$u_{xx} + u_{yy} = u_{\xi\xi} + u_{\eta\eta} = 0 \Rightarrow$ Thus

By (i),

$u \in C^2(\Omega) \Rightarrow u(\xi,\eta) \in C^2(\Omega')$ & by (ii)

$\Delta u(x,y) = 0 \Rightarrow \Delta u(\xi,\eta) = 0$, thus

the transforma-n <u>preserve</u> harmonicity of u.

$\blacksquare$